A complete research tree

Same research order as the sketch: design decisions -> representation choice -> Scheme2 detector study -> diagnostics and transfer.

CENTRAL QUESTION

How can local traffic representations and background combinations support attack detection at low false-alarm rates?

1 ESTABLISH THE DATA

Source-IP view
100s sequence / 10s bucket
Feature support and natural hybrids
Protected global-time-block split
B+A / B+B and donor provenance

→

2 CHOOSE THE REPRESENTATION

Event-union reference: 228D
Scheme1: 137D; add L2 control
Scheme2: 64D vector + intensity
Natural validation + frozen real test
Select the common Scheme2 input

→

3 ESTABLISH A DETECTOR

XGBoost / LSTM / CNN / CNN+LSTM
Same data, weights and metrics
Historical single-seed screen
Matched five-seed temporal study
Retain LSTM as working baseline

RESEARCH BRANCHES FROM THE SHARED SCHEME2 DETECTOR

A TIME AND CONTEXT

Circular shift: small overall AP changes, localized type/threshold effects.
Window and bucket scale: context and coverage change together. [E20-E26]

B ATTACK DIFFICULTY

Web: often isolated local evidence.
Infilteration: input ambiguity, host composition and two mixing profiles.
Geometry + raw-flow audits. [E14-E29]

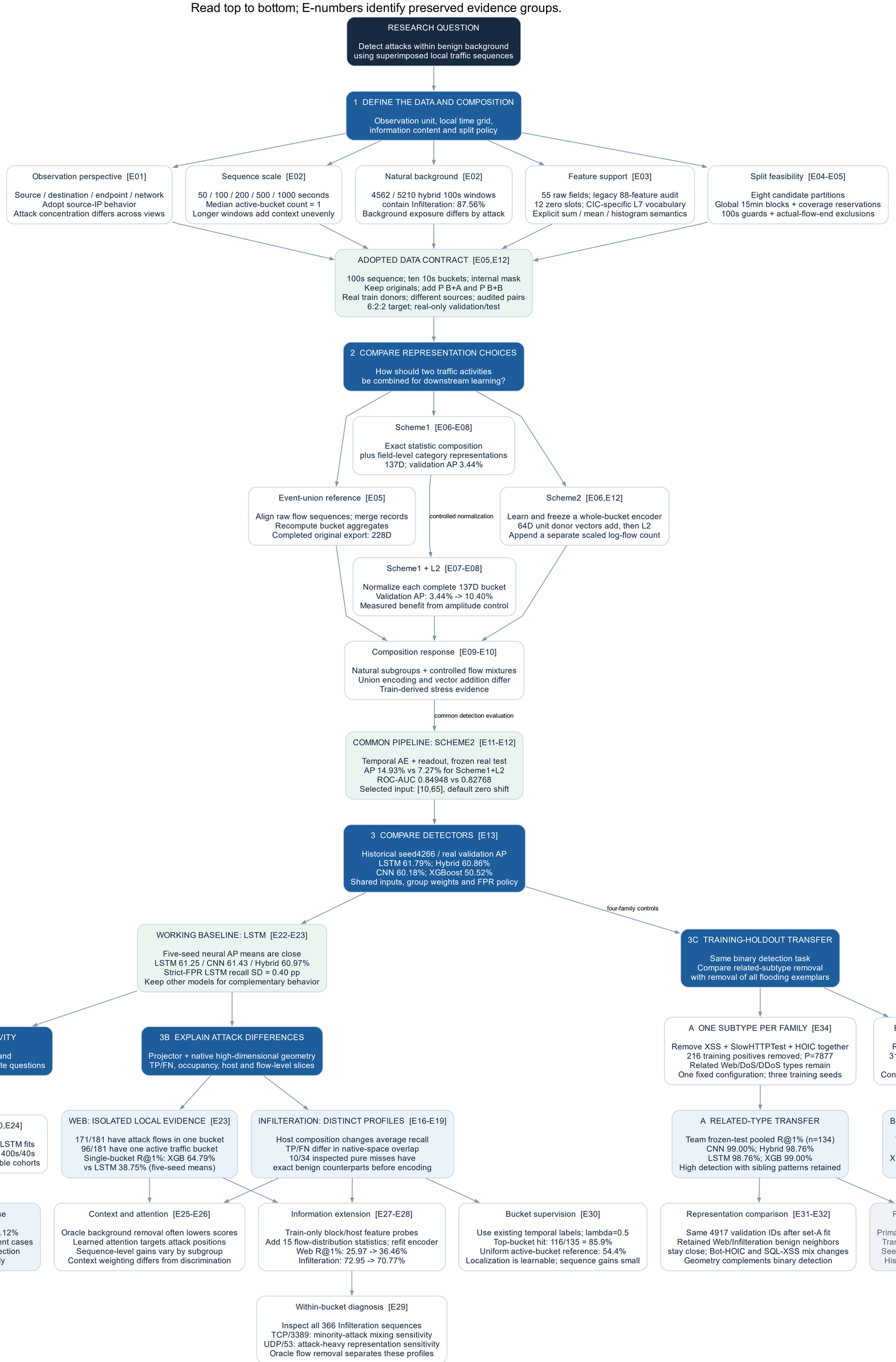
C TRAINING-HOLDOUT TRANSFER

Set A: retained sibling patterns support high pooled held-out recall (~99%).
Set B: flooding removal lowers pooled recall by 26.57-31.68 pp. [E34-E35]

TARGETED EXPERIMENTS CONNECT THE BRANCHES

Context interventions -> learned attention: selective weighting and sequence discrimination are distinct.
Feature-family screening -> 15-statistic extension: similar AP can conceal a Web/Infilteration tradeoff.
Bucket labels -> auxiliary supervision: attack localization is learnable (116/135 Infilteration cases; 85.9%).
Matched-ID embedding views -> transfer analysis: retained-type benign mixing stays close while attack neighborhoods reorganize.

COMPLETE RESEARCH TREE | Research logic, not a timeline | 29 October 2026
Read top to bottom; E-numbers identify preserved evidence groups.



Data design and representation choice

Foundation and comparison branches from the sketch; evidence E01-E12.

DATA DECISIONS FROM FOUR EDA STUDIES

Perspective: compare source / destination / endpoint / network; adopt source-IP behavior.
Scale: compare 50/100/200/500/1000s; adopt 100s plus ten 10s buckets.
Natural background: 87.56% of hybrid 100s windows contain Infilteration.
Features: audit 55 raw fields and the old 88-feature list; use supported CIC-specific inputs.
Split: global 15min blocks, coverage reservations, 100s guards and actual-flow-end exclusions.

COMPOSITION AND FITTING CONTRACT

Retain all real training originals. P counts real attack-bearing TRAINING SEQUENCES, including natural hybrid. Add P B+A and P B+B.
Different-source donors come only from real training originals. Type/source and background-activity sampling are explicit.
Fit vocabularies, scales and input-encoder weights from retained training data; freeze the representation before synthetic export.
Validation and test retain real natural populations.

Representation / temporal AE + readout	Validation AP	Frozen test AP	Meaning
Scheme1 (137D)	3.44%	--	Statistical composition + field representations
Scheme1 + bucket L2	10.40%	7.27%	Controlled amplitude normalization
Scheme2 (64D + count)	19.65%	14.93%	Selected whole-bucket representation

WHAT THE NORMALIZATION COMPARISON ADDS

Scheme1's highest-error 0.1% of validation sequences account for 41.66% of reconstruction error; after L2, 0.80%.
This supports amplitude control as a contributor to improved representation behavior. L2 fixes vector length while preserving relative feature magnitudes within a bucket.
Whole-bucket encoding, field choices and composition rules also differ between Scheme1 and Scheme2.

WHAT THE COMPOSITION COMPARISON ADDS

Raw-flow union and separate-encoding addition generate measurably different representations and decisions. Controlled mixtures reuse training donors and are stress diagnostics.
Scheme2 is selected from completed natural-validation and frozen real-test comparisons.
Unit donor embeddings enter the sum with equal coefficients when both are active; a separate total-flow-count channel preserves combined intensity.

Detector selection and temporal sensitivity

Historical model screen and same-runtime five-seed temporal comparison are distinct baselines.

Historical seed4266 / validation	AP	R at <=1% FPR	R at <=0.1% FPR
LSTM	61.79%	76.34%	50.49%
CNN + LSTM	60.86%	72.96%	51.04%
CNN	60.18%	74.26%	49.07%
XGBoost	50.52%	69.57%	37.62%

WORKING BASELINE WITH A METRIC-SPECIFIC RATIONALE

LSTM leads the initial AP-first screen. In matched five-seed no-shift fits, AP means are close: LSTM 61.25%, CNN 61.43%, Hybrid 60.97%.
LSTM recall SD at the strict 0.1% FPR budget is 0.40 percentage points on this fixed cohort. This supports retaining it as a working baseline; model complementarity remains useful.
CNN+LSTM here is the downstream Hybrid, distinct from an earlier proposed AE+CNN.

CIRCULAR SHIFT: POSITION SENSITIVITY

Rotate all channels and masks to [5,6,7,8,9,0,1,2,3,4]. Train and validation are rotated consistently; models are refitted.
The paired study covers 40 fits: 36 new, four reused; five seeds. Mean AP deltas in XGB/LSTM/CNN/Hybrid are -0.040/-0.076/+0.116/-0.002 pp.
LSTM Web recall at 1% changes 35.58% -> 29.83%; temporal effects are type- and operating-point-specific.

WINDOW LENGTH AND BUCKET RESOLUTION

Separate raw past-context availability from newly trained detectors. Study 200s/10s, 200s/20s, 400s/10s and 400s/40s.
The 400s/40s Web recall at 1% is 58.12%, versus historical 100s/10s 47.51%. Rebuilt cohorts differ: Web n=117 versus 181; Infilteration n=45 versus 366.
This establishes scale-specific sensitivity and coverage effects rather than a paired improvement on identical examples.

READ THE TEMPORAL RESULT AT ITS TESTED LEVEL

A circular rotation preserves most adjacency. This experiment measures adaptation to a different positional grid, rather than invariance of an unchanged fitted model.
A longer window adds useful history for some traffic; many hard isolated Web cases remain isolated.
AP, low-FPR recall, type slices, occupancy and host composition supply complementary evidence about temporal behavior.

Attack differences lead to targeted experiments

Evidence E14-E30: geometry, raw-flow ambiguity, context, information content and temporal supervision.

WEB: LOCAL EVIDENCE AND MODEL COMPLEMENTARITY

171/181 sequences contain attack flows in just one bucket; 96/181 have only one active traffic bucket.
In the single-active-bucket group, five-seed R@1% is 64.79% for XGBoost and 38.75% for LSTM. LSTM is stronger in the observed multi-bucket groups.
Projector views, full-dimensional neighbors and frozen errors support a content-and-readout question, alongside temporal sparsity.

INFILTERATION: AMBIGUITY AND MULTIPLE PROFILES

10 of 34 inspected pure misses have exact benign counterparts before the bucket encoder. Other misses remain architecture-dependent.
Host weighting materially changes the descriptive average. Later flow audits distinguish TCP/3389 minority-attack mixing from UDP/53 attack-heavy representation response.
These are specific observable tradeoffs; pure means attack-labeled flows, not independently validated malicious semantics.

Same-runtime seed4266 variant	AP	Inf R@1 / 0.1%	Web R@1 / 0.1%
Reference	61.5277%	72.95 / 36.61%	25.97 / 3.87%
Attention pooling	61.87%	70.49 / 35.79%	29.28 / 2.21%
+15 distribution statistics	61.4998%	70.77 / 31.69%	36.46 / 6.08%
Bucket labels; lambda=0.5	61.6398%	74.04 / 36.07%	26.52 / 3.87%

INFORMATION AND MIXING

Train-only block/host probes identify small consistent AUC increments from compact distribution/joint statistics. The detector extension includes a new bucket encoder.
The 23 new strict-budget Infilteration misses include 10 TCP/3389 cases (9 recover after oracle benign-flow removal) and 12 UDP/53 cases (1 recovers).
Deleting pure-benign buckets recovers 3/23; deleting every benign flow recovers 11/23: some effects occur within buckets.

LOCALIZATION AND DISCRIMINATION

Attention learns nonuniform emphasis on attack positions, but sequence effects depend on subgroup and threshold.
Bucket supervision uses existing temporal labels. In context-bearing Infilteration, top-bucket hit rate is 116/135 = 85.9%, versus 54.4% under uniform active-bucket selection.
These results distinguish learnable localization from low-FPR sequence discrimination. Oracle interventions are diagnostics, not deployed improvements.

Training-holdout transfer: updated team findings

Committed team protocol 44ef865 | three seeds 4266-4268 | binary detection on the historical frozen test population.

SET A: RELATED TYPES REMAIN

Simultaneously remove XSS, SlowHTTPTest and HOIC primary labels from training. Remove 216 of 8093 positives; P=7877.
Related Web / DoS / DDoS examples remain. The pooled test slice has 134 positives. This is one fixed subtype-removal configuration with repeated training seeds.

SET B: ALL FLOODING EXEMPLARS REMOVED

Remove all seven DoS and DDoS primary labels together. Remove 313 of 8093 positives (3.9%); P=7780.
The pooled test slice has 202 positives. Each arm refits its input transforms/encoder; detector seeds reuse the arm's frozen data and representation.

Model	A: held-out R@1%	A: paired delta (pp)	B: held-out R@1%	B: paired delta (pp)
CNN	99.00 +/- 0.43%	-0.75 +/- 0.75	68.32 +/- 0.00%	-31.68 +/- 0.00
Hybrid	98.76 +/- 1.14%	-1.00 +/- 0.86	68.32 +/- 0.00%	-31.68 +/- 0.00
LSTM	98.76 +/- 1.14%	-0.50 +/- 1.14	72.94 +/- 4.00%	-26.57 +/- 3.14
XGBoost	99.00 +/- 0.43%	+2.74 +/- 0.43	68.65 +/- 0.29%	-27.72 +/- 0.50

WHAT THE REPLICATION ESTABLISHES

Set A retains high pooled recall when related patterns remain. Set B lowers pooled recall in all three seeds for every model.
Transfer differs across flooding patterns: the team's CNN subtype AP is 1.000 for GoldenEye, SlowHTTPTest and HOIC; LOIC-HTTP mean AP is 0.545 and recall 60/110=54.55%; LOIC-UDP AP rounds to 0.000.
Retained-family effects are model-dependent. In set B, XGBoost Web recall changes 36.55 -> 9.04% and Infilteration 52.99 -> 42.81% (3/3 lower); LSTM Web recall changes 37.55 -> 26.31% with nearly unchanged AP.
These patterns support sibling-assisted and traffic-pattern-dependent transfer, with ranking and threshold changes distinguished.

SCOPE AND REPRESENTATION LINK

The dataset excludes held-out PRIMARY labels; non-primary memberships of retained mixed sequences require the build's membership audit.
Validation retains the withheld types and selects checkpoints/thresholds. The test population was already inspected historically. Three seeds quantify detector retraining variation conditional on fixed data and the arm's encoder.
Earlier matched-ID set-A visualization shows similar retained Web/Infilteration benign-neighbor fractions but changed Bot-HOIC and SQL-XSS neighborhoods.
Numbers on this page are reported in the committed team protocol; original three-seed prediction arrays are not present in this checkout and were not re-scored here.

Corrections, additions and evidence

The final tree preserves the sketch's conceptual order while attaching each conclusion to its actual measurement.

PRECISIONS APPLIED TO THE SKETCH

1. Event-union composition aligns raw flow sequences and reaggregates them; bucket statistics are not added indiscriminately.
2. Natural hybrid Infilteration membership is 4562/5210 = 87.56%.
3. P counts attack-bearing real TRAIN sequences, rather than malicious flows.
4. Global 15min blocks define split grouping/isolation; sparsity is a separate EDA finding.
5. L2 controls per-vector amplitude; it preserves relative coordinate magnitudes.
6. Scheme2 adds unit donor vectors with equal coefficients and retains a separate intensity channel.
7. LSTM's low seed variation is specific to strict-FPR recall, rather than every metric.
8. Set-A and set-B findings describe conditional transfer; they do not divide all unseen attacks into a universal can/cannot generalize rule.

BRANCHES ADDED TO MAKE THE TREE COMPLETE

Data: full feature audit, donor balancing, source provenance, coverage reservations and explicit empty-bucket handling.
Representation: amplitude diagnosis, natural subgroup response, controlled flow mixtures and the completed AE/readout real-test baseline.
Detector study: matched seeds, operating-point behavior, raw context availability and visualization with native-dimensional checks.
Hard attacks: host composition, exact benign collisions, TCP/3389 versus UDP/53 and within-bucket mixing.
Interventions: learned attention, compact distribution features and temporal bucket supervision.
Transfer: three-seed subtype and all-flooding removal, retained-family shifts, ranking/threshold distinction and matched-ID geometry.

Evidence groups	Research branch	Evidence scope
E01-E05	Observation, scale, hybrid, feature and split EDA	Descriptive source / completed construction
E06-E12	Representation comparisons and Scheme2 selection	Real validation + historical test + train-derived stress
E13-E19	Model screen, geometry, collisions and labels	Real validation + explicit training-reference reuse
E20-E26	Context, time scale, circular shifts and attention	Validation / raw availability / oracle diagnostics
E27-E30	Feature probes, changed cases and bucket labels	Train-derived probes + exploratory validation
E31-E32	Fixed holdout visualization	Paired-ID validation geometry
E34-E35	Team holdout protocol and seed aggregation	Committed three-seed frozen-test report + code

Full editable tree: docs/research_tree/research_tree.dot and RESEARCH_TREE.md. Local source paths, SHA256 values and the committed team snapshot are indexed under docs/research_map_20261029/. All existing experiment data and weights remain unchanged.